Name: Mimanshu Gahlaut

UID: 24BAI70038

Course: BE-CSE (AI&ML)

Subject: Database Management System

Experiment 9: Packages in PL/SQL

1. Aim of the Session

To design and implement PL/SQL packages by creating both the package specification and package body, incorporating procedures and shared cursors to support modular, reusable, and efficient database operations.

2. Software Requirements:

- Database Management System:
  - Oracle Database Express Edition (Oracle XE)
  - PostgreSQL Database

- Database Administration Tool / Client Tool:
  - Oracle SQL Developer (for Oracle XE)
  - pgAdmin (for PostgreSQL)

3. Objective of the Session

To develop a PL/SQL package that includes procedures and shared cursors for organized and modular programming.

4. Practical / Experiment Steps

The work was carried out through the following activities:

1. **Design of Modular Structure:** Divided the database logic into a package consisting of a specification (interface) and a body (implementation).
2. **Shared Cursor Definition:** Created a common cursor within the package body to allow multiple procedures to access the same dataset efficiently.
3. **Procedure Implementation:** Developed procedures for handling different operations such as displaying all records and retrieving specific entries.

4. **Encapsulation Mechanism:** Restricted direct access to internal logic by exposing only the required procedures through the package interface.
5. **Testing and Validation:** Executed the procedures within anonymous blocks to ensure correct functionality and consistency.

5. Procedure of the Practical

Execution was performed in the following order:

• Connected to the database environment and enabled output display for verification.

• Created the employees table and inserted sample data for testing purposes.

• Defined the package specification (`emp_package`) to declare the available procedures.

• Implemented the package body containing the shared cursor and the logic for each procedure.

• Used a loop structure in the `show_employees` procedure to iterate through all records using the cursor.

• Implemented filtering logic in the `get_employee` procedure to retrieve details based on a given employee ID.

• Compiled both the specification and body to ensure error-free execution.

• Executed `show_employees` to display all records.

• Called `get_employee(2)` to verify retrieval of a specific record.

6. I/O Analysis (Input / Output Analysis)

Input Queries

SQL

```sql
CREATE TABLE emp_data (
    id NUMBER PRIMARY KEY,
    name VARCHAR2(50),
    salary NUMBER
);

/
```

```
SQL> CREATE TABLE emp_data (
        id NUMBER PRIMARY KEY,
        name VARCHAR2(50),
        salary NUMBER...
Show more...


Table EMP_DATA created.

Elapsed: 00:00:00.017
```

```
INSERT INTO emp_data VALUES (1, 'Amit', 30000);
INSERT INTO emp_data VALUES (2, 'Riya', 40000);
INSERT INTO emp_data VALUES (3, 'John', 50000);

COMMIT;

/
```

```
SQL> INSERT INTO emp_data VALUES (1, 'Amit', 30000)


1 row inserted.

Elapsed: 00:00:00.015

SQL> INSERT INTO emp_data VALUES (2, 'Riya', 40000)


1 row inserted.

Elapsed: 00:00:00.002

SQL> INSERT INTO emp_data VALUES (3, 'John', 50000)


1 row inserted.

Elapsed: 00:00:00.002

SQL> COMMIT


Commit complete.

Elapsed: 00:00:00.002
```

```sql
CREATE OR REPLACE PACKAGE employee_pkg AS

    -- Display all employee records
    PROCEDURE display_all;

    -- Fetch employee by ID
    PROCEDURE find_employee(p_emp_id NUMBER);

END employee_pkg;

/
```

```
SQL> CREATE OR REPLACE PACKAGE employee_pkg AS

        -- Display all employee records
        PROCEDURE display_all;...
Show more...


Package EMPLOYEE_PKG compiled

Elapsed: 00:00:00.011
```

```sql
CREATE OR REPLACE PACKAGE BODY employee_pkg AS

    -- Shared cursor
    CURSOR emp_cur IS
        SELECT id, name, salary FROM emp_data;

    -- Procedure to display all records
    PROCEDURE display_all IS
    BEGIN
        FOR rec IN emp_cur LOOP
            DBMS_OUTPUT.PUT_LINE(
                'ID: ' || rec.id ||
                ', Name: ' || rec.name ||
                ', Salary: ' || rec.salary
            );
        END LOOP;
    END;

    -- Procedure to fetch specific employee
    PROCEDURE find_employee(p_emp_id NUMBER) IS
    BEGIN
        FOR rec IN emp_cur LOOP
            IF rec.id = p_emp_id THEN
                DBMS_OUTPUT.PUT_LINE(
                    'Employee Details -> ID: ' || rec.id ||
                    ', Name: ' || rec.name ||
                    ', Salary: ' || rec.salary
                );
            END IF;
        END LOOP;
    END;
```

```
END employee_pkg;

/
```

```
SQL> CREATE OR REPLACE PACKAGE BODY employee_pkg AS

         -- Shared cursor
         CURSOR emp_cur IS...
Show more...


Package Body EMPLOYEE_PKG compiled

Elapsed: 00:00:00.032
```

```
SET SERVEROUTPUT ON;

BEGIN
    employee_pkg.display_all;
END;

/
```

```
SQL> BEGIN
         employee_pkg.display_all;
     END;


ID: 1, Name: Amit, Salary: 30000
ID: 2, Name: Riya, Salary: 40000
ID: 3, Name: John, Salary: 50000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.126
```

```
BEGIN
    employee_pkg.find_employee(2);
END;

/
```

```
SQL> BEGIN
         employee_pkg.find_employee(2);
     END;


Employee Details -> ID: 2, Name: Riya, Salary: 40000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005
```

7. Learning Outcome

- **Understanding Package Structure:** Learned the distinction between package specification and package body.
- **Improved Code Organization:** Gained knowledge of grouping related operations into a single unit for better maintainability.
- **Efficient Data Handling:** Understood how shared cursors help in reducing redundancy and improving performance.
- **Encapsulation Concept:** Learned how packages hide internal implementation details and provide a clean interface.